TEXORA · LECTURE SHEET

Fabric Strength Tests: Tensile, Tearing, Ballistic and Bursting, Abrasion Resistance

A fabric can fail by being pulled apart, torn from a nick, burst from inside, or worn away thread by thread — and each failure has its own test. This sheet covers strip and grab tensile testing, Elmendorf and related tear methods, bursting strength with diaphragm correction, and abrasion as a strength-loss endpoint.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 3 · LEVEL 3 TERM 2

1. Four ways a fabric fails

Strength is not one property. In service a curtain fails at its hem under a steady pull, a trouser seat fails by a tear spreading from a small cut, an upholstery fabric fails by bursting over a frame edge, and a sock fails by abrasion long before its unused yarn strength is exhausted. Each mode loads the structure differently: tensile testing loads all threads in a strip together, tearing concentrates load on one or two threads at a tear tip, bursting loads the fabric biaxially over an area, and abrasion removes material until the remaining threads can no longer carry ordinary loads. A complete strength profile therefore needs at least one test from each family, and a good result in one family does not guarantee a good result in another.

All tests in this sheet are made on specimens conditioned at 65% RH and 20 °C, cut away from selvedges and visible faults, with warp and weft directions tested separately for woven fabrics. Results are reported as means of at least five valid specimens per direction, with the direction, specimen geometry and standard method stated.

2. Tensile strength — strip and grab methods

2.1 The ravelled strip method (ISO 13934-1)

In the strip method the specimen is a strip of fabric whose width is fully gripped by both jaws. The standard working width is 50 mm: a wider strip is cut and then ravelled — threads are removed from both edges — until exactly 50 mm of full threads remains, so that no edge thread is a partial thread. The strip is mounted in a constant-rate-of-extension (CRE) tester, typically with a 200 mm gauge length for

woven fabrics, and extended until it breaks. The maximum force is the breaking strength for that direction; extension at break is recorded from the same test.

Ravelling matters. A strip cut to width with scissors leaves damaged and partial edge threads that fail early and scatter the results; a ravelled strip presents a known number of complete threads to the jaws — the very quantity used in the fabric-assistance calculation of Topic 4.

2.2 The grab method (ISO 13934-2)

In the grab method a wider specimen, commonly 100 mm, is gripped only in its central part by narrower jaws, commonly 25 mm wide. Only the threads directly between the jaws are fully loaded, but threads near the jaw edges receive partial assistance from their neighbours, so the result is an effective strength for a defined grip geometry rather than the strength of a counted number of threads. The grab test is faster — no ravelling — and it imitates many service grips, where fabric is held locally (a seam, a strap, a buttoned edge) rather than across its whole width. Its results are not directly interchangeable with strip results, and the two methods must never be mixed in one specification.

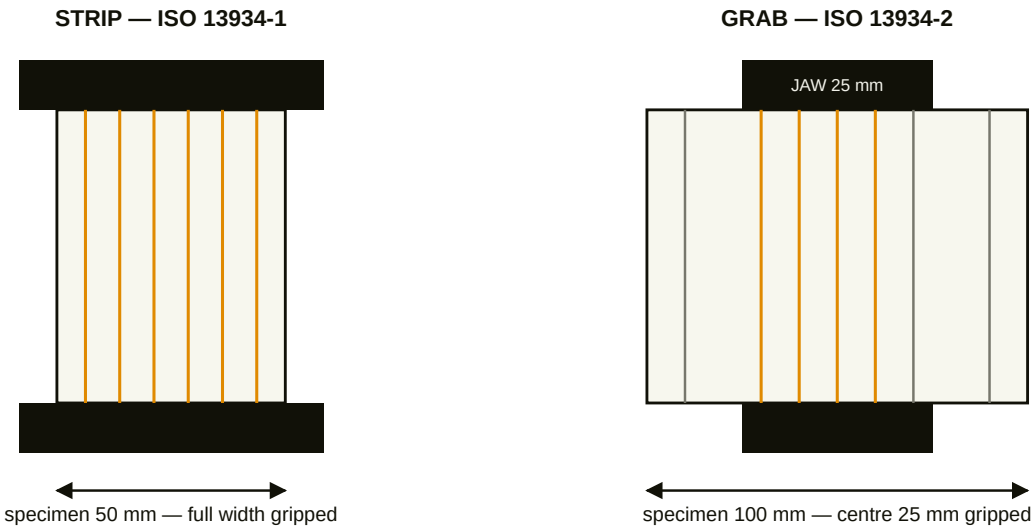


Figure 1 — Strip versus grab specimens. In the strip test the whole 50 mm width is gripped; in the grab test a 100 mm specimen is gripped only over its central 25 mm.

FEATURE	STRIP (ISO 13934-1)	GRAB (ISO 13934-2)
Specimen	50 mm, ravelled to exact width	100 mm wide, centre gripped by 25 mm jaws
Threads loaded	Known, countable number	Central threads fully, neighbours partially
Preparation	Slow — ravelling required	Fast — cut only

Best use	Specification, research, fabric-assistance analysis	Routine control, local-grip service simulation

WORKED EXAMPLE — STRIP TENSILE STRENGTH

Five warp-way ravelled strips, each 50 mm wide, broke at 412, 428, 405, 435 and 420 N.
Sum = 412 + 428 + 405 + 435 + 420 = 2,100 N. Mean warp breaking strength = 2,100 / 5 = **420 N**.
Expressed per unit width: 420 N / 5 cm = **84 N/cm**.
The weft-way mean for the same fabric was **324 N**. The fabric is warp-stronger, as expected from its higher warp density (70 EPI against 59 PPI, Topic 1) and its higher warp crimp history — a reminder that a single strength number without a direction is meaningless.

3. Tearing strength

A tear starts at a cut, nick or weak place and spreads thread by thread. Because the load is concentrated at the tear tip, tear strength depends less on the total number of threads than on whether threads can move, bunch together and share the load at the tip. A loose weave that lets threads slide into a bunch can out-tear a tighter fabric of stronger yarn — the opposite of tensile behaviour — which is why tear must be tested, not inferred.

3.1 Elmendorf ballistic pendulum (ISO 13937-1)

The Elmendorf tester is a pendulum carrying a clamp. The specimen is held between a fixed clamp and the pendulum clamp, a slit is cut to start the tear, and the pendulum is released. The energy the pendulum loses in tearing the specimen through a fixed distance is read from its swing and converted to a tearing force. The method is rapid and is the common specification test for apparel and many industrial fabrics; it suits light to medium fabrics, while very strong fabrics exceed the pendulum capacities and need a tensile-machine tear method.

WORKED EXAMPLE — ELMENDORF TEAR RESULT

A set of warp-way Elmendorf specimens gave a mean tearing force of **14.1 N**.
Reading the result: this is the mean force associated with propagating the tear over the standard tearing distance, derived from the pendulum energy loss. It is reported as a mean of the valid specimens in each direction, with the pendulum capacity stated, because a result taken near the top or bottom of a pendulum scale is less reliable than one taken in its middle range.

3.2 Tongue and trouser tear — in outline

In the **tongue tear** method a rectangular specimen is cut to form two tongues, the tongues are gripped in the two jaws of a CRE tester and pulled apart, and the force trace shows a series of peaks as successive threads break; a mean of the defined peaks is reported. In the **trouser tear** method the specimen is cut like a pair of trousers and the legs are pulled apart in the same way. Both methods tear at a controlled rate on a tensile machine, handle strong fabrics beyond pendulum capacity, and produce a force–extension record that reveals whether tearing is steady or stick-slip. Their results are method-specific and are not numerically interchangeable with Elmendorf values.

4. Bursting strength

Bursting tests load the fabric in all directions at once, so they are the natural test for knits — which have no single warp and weft — and for any fabric that fails in service under biaxial pressure. In the hydraulic and pneumatic methods of ISO 13938-1 and ISO 13938-2, the specimen is clamped as a diaphragm over a circular opening, a rubber diaphragm beneath it is inflated by fluid pressure (hydraulic) or air pressure (pneumatic), and the pressure at which the fabric bursts is recorded, together with the distension (height of the dome) at burst where required.

4.1 Diaphragm correction

Part of the recorded pressure is spent stretching the rubber diaphragm itself, not the fabric. The pressure needed to distend the diaphragm alone to the same height, measured without a specimen, is therefore subtracted:

$$\text{Bursting strength} = \text{total pressure at burst} - \text{diaphragm correction pressure}$$

WORKED EXAMPLE — BURSTING STRENGTH

A specimen burst at a total pressure of 320 kPa. The diaphragm alone required 28 kPa to reach the same distension.

Bursting strength = 320 − 28 = **292 kPa**.

Omitting the correction would overstate the fabric by almost 10% — and the error grows in proportion for weaker fabrics, where the correction is a larger fraction of the total.

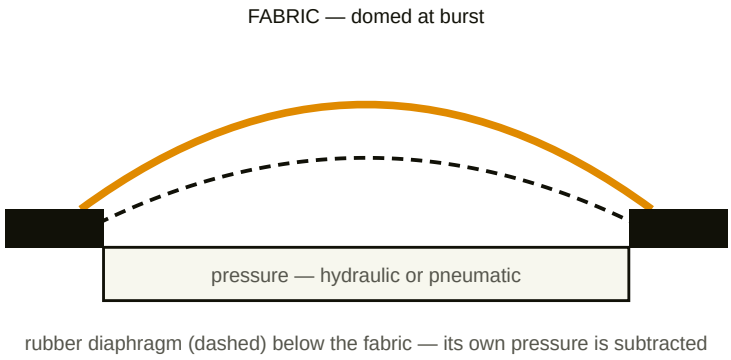


Figure 2 — Bursting principle. The clamped fabric is domed by pressure acting through a rubber diaphragm until it bursts; the diaphragm correction is deducted from the total pressure.

4.2 Ball-burst — in outline

In the ball-burst method a polished steel ball is pushed through the clamped fabric by a compression cage on a tensile tester, and the force at rupture is recorded. It needs no fluid system and handles strong fabrics, but it reports a force rather than a pressure and its geometry differs from the diaphragm methods, so ball-burst and diaphragm-burst values are kept in separate specifications.

5. Abrasion resistance as a strength-loss endpoint

Abrasion is the wearing away of fabric by rubbing. It is treated fully in a later topic; here it enters as a strength story. An abraded fabric rarely fails because its surface looks worn — it fails because so much material has been removed that the remaining threads break under ordinary load. Two endpoints are therefore used in abrasion testing: a visual endpoint (a stated number of threads broken, or a defined change of appearance) and a strength endpoint (specimens are abraded for stated cycles on a Martindale tester, in which the specimen rubs against a standard abradant in a multidirectional Lissajous motion under a stated pressure, and the residual tensile or bursting strength is then measured). Reporting strength loss after a fixed number of cycles — for example the percentage of original strip strength retained — links this topic directly to end use: a fabric that keeps 90% of its strength after the specified abrasion will outlast one that keeps 50%, whatever their unused strengths were.

6. Choosing and reporting strength tests

FAILURE MODE	TEST	RESULT UNIT	WATCH POINT
Pull apart	Strip / grab tensile	N (per stated width)	Direction; method not interchangeable

Tear spread	Elmendorf; tongue; trouser	N	Thread mobility at the tear tip
Biaxial burst	Hydraulic / pneumatic diaphragm	kPa, after correction	Diaphragm correction always deducted
Wear through	Martindale abrasion + residual strength	Cycles; % strength retained	State the endpoint used

Every report states: method and standard, direction, number of specimens, mean result, atmosphere (65% RH, 20 °C), and any correction applied. A strength value without those companions cannot be checked, compared or specified.

Glossary

Ravelled strip — tensile specimen reduced by removing edge threads to an exact width of complete threads. **Grab test** — tensile test in which only the centre of a wider specimen is gripped. **Elmendorf tester** — ballistic pendulum tear tester. **Tongue tear** — CRE tear test on a specimen cut into two tongues. **Bursting strength** — pressure at rupture under biaxial loading, after diaphragm correction. **Diaphragm correction** — pressure needed to distend the rubber diaphragm alone, deducted from the total. **Distension** — dome height of the specimen at burst. **Martindale tester** — abrasion tester using a Lissajous rubbing motion.

Key points

• Tensile, tear, bursting and abrasion measure different failure modes; no one of them predicts the others.
• Strip testing (ISO 13934-1) loads a counted 50 mm width; grab testing (ISO 13934-2) grips the centre of a 100 mm specimen — results are not interchangeable.
• Elmendorf tearing (ISO 13937-1) reads pendulum energy loss as a tearing force; tongue and trouser methods cover stronger fabrics on a CRE machine.
• Bursting strength (ISO 13938) is always reported after subtracting the diaphragm correction.
• Abrasion is usefully reported as residual strength after stated cycles, not only as appearance change.

Practice

1. Five weft-way strip specimens break at 310, 335, 318, 330 and 327 N. Calculate the mean breaking strength and express it in N/cm for a 50 mm strip.
2. A fabric bursts at 410 kPa total pressure and the diaphragm correction is 35 kPa. Calculate the bursting strength and the percentage by which the uncorrected value would overstate it.

3. Explain why a tightly woven fabric of strong yarn can show a lower Elmendorf tear strength than a looser fabric of weaker yarn.
4. A specification simply says tensile strength 400 N. List four pieces of information that must be added before the requirement can be tested fairly.

Answer key

1. Sum = 310 + 335 + 318 + 330 + 327 = 1,620 N. Mean = 1,620 / 5 = **324 N**. Per unit width: 324 / 5 = **64.8 N/cm**.
2. Bursting strength = 410 − 35 = **375 kPa**. Overstatement = 35 / 375 × 100 = **9.3%**.
3. Tearing loads only the threads at the tear tip. In a tight fabric those threads cannot move, so one thread takes the load, breaks, and the tear advances thread by thread at low force. In a looser fabric threads slide and bunch at the tip, sharing the load among several threads at once, so a higher force is needed to propagate the tear even though the individual yarns are weaker.
4. Any four of: test method (strip or grab) and its standard; direction (warp or weft); specimen width and gauge length; number of specimens and whether 400 N is a mean or a minimum; conditioning atmosphere. Without method and direction the same fabric can both pass and fail the same written number.
